



Atlas 5 launch delayed

The next launch of the Atlas 5 has been delayed to no sooner than July 9 due to a failed battery. <https://digit.in/july19-18>



ISRO breaks world record

ISRO created a new world record by sending 104 satellites into Earth's orbit, which beats out NASA as well. <https://digit.in/july19-19>

forms of life could breathe in hydrogen and breathe out methane. They can consume acetylene and ethane to produce energy. In the subsurface liquid oceans, conditions are suitable to harbour life as it is on Earth. Life may not even be tied to a planet at all. Dust suspended in plasma can get charged and form helical structures. Computer simulations have shown that these structures are capable of reproducing. An advanced species may have uploaded itself entirely into computers, transcending physical bodies, living in a post-biological state. We may end up meeting ET in the cloud.

The universe is apparently teeming with life. Yet, we have absolutely no evidence of it. Actually, there is no widely-acknowledged existence of extraterrestrial life. We might have already found life in almost every place we looked. The Venera probe on Venus photographed slug like creatures that apparently move between photographs. The Viking landers on Mars may have detected signs of organic chemistry. The amount of methane in the martian atmosphere waxes and wanes, it is unclear if the process behind it is biological or geological. On Titan, the levels of acetylene and ethane are less than expected, consistent with the existence of an exotic form of life. However, none of these are considered unequivocal proofs of the existence of ET or advanced civilisations. As far as humans go, we are alone in the universe. That is the essence of the question asked by physicist Enrico Fermi. The galaxy should be full of advanced extraterrestrials. Where are they?

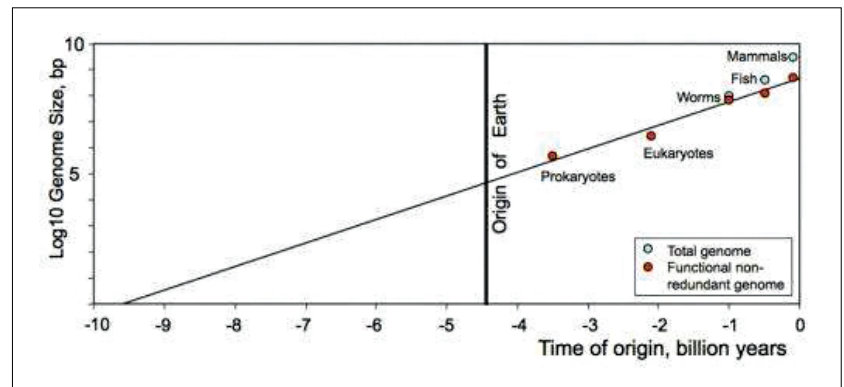
THE SOLUTIONS

This is a mystery with few solutions, really. UFO conspiracy theorists believe that world governments are intentionally hiding the evidence of aliens from the world. Yet, there isn't a single verified photo, not even a speck of alien dust available to researchers to confirm these allegations. Despite the overwhelming number of fakes, dedicated researchers are still willing to

scan every shred of evidence. It has not paid off so far, but we can rest assured that even if some unlucky person gets tractorated up to a flying saucer and probed by aliens, we will all know. The number of serious possible solutions to the Fermi Paradox boils down to five main hypotheses. Two of these, the Rare Earth hypothesis and the Great Filter, deal with the trajectory of the growth and development of the human civilisation. The remaining, the Zoo, Planetarium and Dark Forest hypotheses suggest that the aliens are actively preventing Earth from finding out about their existence.

being the only species that has gone through all the stages. The Great Filter may be ahead or behind us, but in either case, it makes the detection of alien life an extremely unlikely event.

According to the Zoo hypothesis, advanced aliens are aware of humans on Earth, and are deliberately avoiding contact, to prevent possible disturbance to our evolution and cultural development. There is also the possibility of introducing diseases, if direct contact occurs. The aliens may simply be risk averse, and are hiding themselves for that reason. According to the planetarium hypothesis, the uni-



A graph mapping the increasing complexity of life on Earth

According to the Rare Earth hypothesis, too many factors went exactly right on Earth, for life to emerge. Just being in the hab zone is not enough, a gas giant such as Jupiter is also necessary to guard Earth from too many asteroid impacts. We do not know what led to the emergence of the eukaryotic cell, and that may be an uncommon event in the universe. The diversification of life during the Cambrian explosion could also be rare, and a technologically advanced species like humans may simply not emerge. According to the Great Filter theory, a civilisation that can be detected by man has to emerge around the right star, have a self replicating molecule such as DNA, must pass through the stage of single cell life, evolve sexual reproduction, begin using tools, and begin to colonise the galaxy. At least one of these steps may be extremely rare, and we have no way of telling,

verse around us is an illusion created by an advanced civilisation, to prevent humans from ever finding aliens. The Dark Forest hypothesis by the Chinese science fiction author Liu Cixin is perhaps the most terrifying of all explanations. Every time a civilisation has met another on Earth, it has been eradicated. There were six species of Homo at various points on Earth, but sapiens are the only one that survived. A civilisation can also be infected by ideas or code. Therefore, all civilisations keep a low profile, to prevent extermination by galactic superpredators out there. If you see everyone running or hiding, the smart thing would be to also run or hide. The Fermi Paradox can cause the Fermi Paradox. Lying low is the wisdom of the silent stars. While other intelligent advanced civilisations have adopted the approach of concealing themselves, humans are yet to learn this valuable lesson.